



Original article

Low visceral fat volume and hypoalbuminemia as prognostic markers in hospitalized patients with coronavirus disease 2019 during the omicron variant epidemic

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ARTICLE INFO

Article history:

Received 24 May 2024

Accepted 22 September 2024

Keywords:

COVID-19

Omicron

Visceral fat area

Serum albumin

SUMMARY

Background & aims: The rate of severe cases of coronavirus disease 2019 (COVID-19) has decreased since the Omicron variant became epidemic. Visceral fat volume was a risk factor for COVID-19 severity with prior prevalent variants, but whether visceral fat volume remains a risk factor for the Omicron variant is unclear. We investigated the associations of clinical factors including visceral fat volume with severity and mortality among hospitalized patients with COVID-19 during the Omicron variant epidemic.

Methods: This was a single-center retrospective cohort study conducted at the Teikyo University Hospital in Japan. We included hospitalized patients with COVID-19 during the Omicron variant epidemic who underwent computed tomography of the abdomen. Clinical data were obtained from the medical records and visceral fat area (VFA) was measured using a 3-dimensional image analysis system volume analyzer. Severity was determined by the presence or absence of oxygen supplementation.

Results: Among the 226 patients, 66 patients showed moderate severity and 29 patients were non-survivors. Hypoalbuminemia was associated with severity (odds ratio [OR] 3.93, 95 % confidence interval [CI] 1.91–8.07; $p = 0.0002$), and hypoalbuminemia (OR 8.38, 95%CI 2.37–29.58; $p = 0.0010$) and low VFA (OR 3.40, 95%CI 1.15–10.06; $p = 0.027$) were associated with mortality. Decision tree analysis showed that mortality rate in the hypoalbuminemia and low-VFA group (37.3 %) was significantly higher than in other groups ($p \leq 0.01$).

Conclusions: Low visceral fat volume and hypoalbuminemia were associated with mortality in hospitalized patients with COVID-19 during the Omicron variant epidemic. Classification by VFA and serum albumin may allow simple prediction of mortality risk among hospitalized patients with COVID-19.

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1. Introduction

An outbreak of coronavirus disease 2019 (COVID-19) was recognized in late December 2019, caused by infection with severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2) [1]. The World Health Organization declared a epidemic on March 11, 2020 [2]. Increasingly more infectious variants emerged during the global epidemic, causing waves of infections across countries and regions [2]. In November 2021, the World Health Organization

identified the B.1.1.529 variant and named it the Omicron variant [3]. This variant subsequently became a major variant in subsequent waves of the epidemic. Compared to previous variants, the Omicron variant saw a reduction in the percentage of COVID-19 patients requiring oxygen therapy from 74 % to 17.6 % and a decrease in the mortality rate from 29.1 % to 2.7 % [4].

The severity of COVID-19 has been reported to involve various factors, including underlying health conditions, lifestyle, and age. Obesity was reportedly associated with COVID-19 severity and mortality prior to the emergence and prevalence of the Omicron variant [5]. Visceral fat volume was also reported as a risk factor for severity [5], with severe patients having a visceral fat area (VFA) on average 41.7 cm² larger than that in non-severe patients [6]. Since

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the emergence of the Omicron variant, reports on VFA and risk of COVID-19 severity or death have been lacking. However, a study of COVID-19 patients background in waves with different major variants showed differences in the frequency of comorbidities [7]. In addition, in a study before the Omicron variant, some factors that were a risk for severity in the previous variants were not a risk for severity in the other variants [4]. Based on these reports, we considered the possibility that visceral fat volume might be changed as a risk for severity or mortality in the Omicron epidemic.

We hypothesized that with the Omicron variant, visceral fat volume may not play a role in severity and mortality among hospitalized patients with COVID-19. The objectives of this study were: first, to analyze the association between severity of COVID-19 in the Omicron epidemic and clinical factors including visceral fat volume; second, to analyze associations between outcome of COVID-19 in the Omicron epidemic and clinical factors including visceral fat volume; and third, to analyze whether a combination of clinical factors and visceral fat volume can be used to assess the risk of mortality.

2. Materials and methods

2.1. Patients

This was a single-center retrospective cohort study conducted at the Teikyo University Hospital in Japan. Participants in this study comprised patients ≥ 20 years old who had been admitted to our hospital between 1 January and 5 August 2022 and had a diagnosis of COVID-19 either on admission or during hospitalization and who underwent computed tomography (CT) of the abdomen. COVID-19 was diagnosed based on a positive result on either a SARS-CoV-2 polymerase chain reaction test or a SARS-CoV-2 antigen test. The SARS-CoV-2 variant identification was conducted using the SARS-CoV-2 Direct Detection RT-qPCR Core Kit, Primer/Probe E484A (SARS-CoV-2), Primer/Probe G339D (SARS-CoV-2), Primer/Probe ins 214 EPE (SARS-CoV-2), and Primer/Probe L452R (SARS-CoV-2) Ver.2 (Takara Bio Inc., Shiga, Japan) in accordance with the instructions provided in the accompanying documentation. Patients who were found to be infected with non-Omicron variants were excluded based on their medical records. Patients who had not undergone abdominal CT and patients for whom VFA could not be measured within the imaging range were excluded. Regarding participant consent, an opt-out format was adopted by the institutional ethics committee. When obtaining information of patients from medical records, etc., information about the research, information about the research, including the purpose of use of the information, were notified, and disclosed to the patients through opt-out on the website. The opportunity for the patients to refuse the use of their information is guaranteed. Patients who did not consent to participate in this the study were also excluded. This study will be carried out according to the principles of the Declaration of Helsinki and Ethical Guidelines for Medical and Health Research Involving Human Subjects published by the Ministry of Health, Labour and Welfare of Japan and the Japanese Ministry of Education, Culture, Sports, Science and Technology. This study received approval from the institutional clinical ethics committee (Teirin 20-030-2).

2.2. CT imaging and measurement of VFA

CT examinations were performed using an Aquilion™ 64 or Aquilion ONE system (Toshiba Medical Systems Corporation, Tochigi, Japan). Images were acquired at collimations of 32×1 mm and 40×1 mm. VFA was analyzed at the umbilical level using the

abdominal analysis (2D) function of the 3D image analysis system volume analyzer (SYNAPSE VINCENT; FUJIFILM Holdings Corporation, Tokyo, Japan).

2.3. Data collection and determination of COVID-19 severity and mortality

The following patient data were extracted from medical records: background (including age and sex), laboratory data on admission (including serum albumin, hemoglobin A1c, and estimated glomerular filtration rate [eGFR]); use of drugs including antiviral agents remdesivir, dexamethasone, sotrovimab, and molnupiravir, and outcome. Comorbidities were defined as follows: Hypertension, dyslipidemia, coronary artery disease, and heart failure in patients on medical treatment; chronic obstructive pulmonary disease and bronchial asthma in patients on inhalation treatment; dialysis in patients on maintenance dialysis; and solid tumors, lymphomas, or leukemia under treatment or follow-up. We accessed medical records between 1 April and 31 August 2023, and after all data were fully anonymized, we did not access to information that could identify individual participants.

In setting cutoffs, the cutoff value for VFA was set at 100 cm^2 . Although no report has established a cutoff for VFA in relation to the risk of COVID-19 severity, an increased risk of obesity-related cardiovascular diseases were reported that in Japanese patients with VFA greater than 100 cm^2 [8]. This conforms to a study before the Omicron variant epidemic, which reported that the patients with VFA of 100 cm^2 or more had a higher rate of severe disease than those with VFA of less than 100 cm^2 [9]. The cutoff value for serum albumin level was set at 3.5 g/dL . This was in accordance with studies before the Omicron variant epidemic, which reported that COVID-19 patients with serum albumin levels less than 3.5 g/dL had higher rates of severe disease and mortality than those with serum albumin of 3.5 g/dL or higher [10,11].

The severity of COVID-19 was determined based on medical records of the admission or diagnosis according to the severity based on Japanese National COVID-19 guidelines, as follows: mild cases, no oxygen therapy required; moderate cases, oxygen therapy required; and severe cases, admission to the intensive care unit or requirement for mechanical ventilation or high-flow oxygen therapy [12]. Non-survivors were defined so as to include deaths due to causes other than COVID-19.

2.4. Statistical analysis

COVID-19 severity was compared between the mild group and the combined moderate and severe groups (moderate-severe group). In terms of outcomes, patients were classified as survivors or non-survivors. Continuous variables were compared using the Mann–Whitney U test for non-parametric data, and categorical variables were compared using the χ^2 test. Multiple logistic regression analysis was conducted to assess factors associated with outcomes and severity. Values of $p < 0.05$ were considered statistically significant in all statistical analyses. All statistical analyses were performed using JMP Pro version 17 (SAS Institute Japan, Tokyo, Japan).

3. Results

3.1. Patient characteristics, COVID-19 severity, and COVID-19 outcome

During the study period, 304 patients were admitted to the hospital with COVID-19 or developed COVID-19 during their

hospital stay. Sixty-seven patients did not undergo abdominal CT on admission or at the time of COVID-19 onset, VFA could not be measured within the imaging range of the abdominal CT in 1 patient, 2 patients were infected with a non-omicron variant (Delta variant), and 8 patients under 20 years of age were excluded because consent for participation in the study could not be obtained. In all, 226 patients were eligible for analysis.

Background characteristics, laboratory data, COVID-19 severity, treatment, and outcomes for the eligible patients are shown in Table 1. Median age was 73.5 years, and 143 (63.3 %) patients were ≥ 65 years of age. Most patients ($n = 133$; 58.9 %) were men and the vast majority ($n = 193$; 85.8 %) of patients had some form of comorbidity. Hypertension was the most common background condition, and solid cancer was present in 50 patients. Median serum albumin was 3.7 g/dL, with serum albumin < 3.5 g/dL in 40.7 %. COVID-19 severity was mild in 160 patients, moderate in 36, and severe in 30. Treatment with COVID-19 drugs was provided for 185 patients. There were 197 survivors and 29 non-survivors.

Median VFA was 108.4 cm² (range, 5.4–354.5 cm²), and VFA was < 100 cm² in 46.5 % of all patients (Table 1). The distribution of VFA is shown in Fig. 1, with the most patients seen in the strata of 50–75 cm² and 100–125 cm² (36 patients each).

Table 1

Clinical characteristics, severity, and mortality of hospitalized patients with COVID-19.

Clinical factor	No. of patients (n = 226)	%
Age (years)^a	73.5	22–97
Age ≥ 65 years (n,%)	143	(63.3)
Male (n,%)	133	(58.9)
Comorbidities (n,%)		
Hypertension	119	(52.7)
Dyslipidemia	72	(31.9)
Coronary artery disease or heart failure	41	(18.1)
COPD or bronchial asthma	21	(9.3)
Solid tumors	50	(22.1)
Lymphomas and leukemia	12	(5.3)
Hemodialysis	8	(3.5)
Laboratory data		
Serum albumin (g/dL) ^a	3.7	1.4–5.3
Serum albumin < 3.5 g/dL (n,%)	92	(40.7)
Hemoglobin A1c (%) ^a	6.0	4.6–14.2
Hemoglobin A1c ≥ 6.5 % (n, %)	55/211	(26.1)
eGFR (mL/min/1.73 m ²) ^a	57.2	3.0–136.3
eGFR < 60 mL/min/1.73 m ² (n, %)	127	(56.2)
Visceral fat area (cm ²) ^a	108.4	5.4–354.5
Visceral fat area < 100 cm ² (n,%)	105	(46.5)
Severity (n,%)		
Mild	160	(70.8)
Moderate	36	(15.9)
Severe	30	(13.3)
Medical treatment (n,%)		
Treated	185	(81.9)
Remdesivir only	91	(40.3)
Dexamethasone only	7	(3.1)
Remdesivir and dexamethasone	52	(23.0)
Other ^b	35	(15.4)
Not treated	41	(18.1)
Outcome (n,%)		
Survival	197	(87.2)
Non-survival	29	(12.8)

Abbreviations: COPD, chronic obstructive pulmonary disease; eGFR, estimated glomerular filtration rate. Data represent the number of patients, with percentages in parentheses.

^a Age, serum albumin, hemoglobin A1c, eGFR, and visceral fat area are given as median and range for each value.

^b Other treatment included sotrovimab in 20 patients, molnupiravir in 12 patients, remdesivir and molnupiravir in 2 patients, and remdesivir and sotrovimab in 1 patient.

3.2. Associations between clinical factors and COVID-19 severity

To analyze associations between clinical factors and COVID-19 severity, we compared clinical factors between the mild group and the moderate-severe group (Table 2). The moderate-severe group showed significantly higher percentages of patients ≥ 65 years of age ($p = 0.013$), male patients ($p = 0.033$), patients with serum albumin < 3.5 g/dL ($p < 0.0001$), and those with hemoglobin A1c > 6.5 % ($p = 0.0018$). Median VFA did not differ significantly between the two groups.

Based on the results of univariate analyses, logistic regression analysis was performed for the association between severity of COVID-19 and each factor (Table 3). Factors associated with severity of COVID-19 were hypoalbuminemia (OR 3.93, 95%CI 1.91–8.07; $p = 0.0002$) and high hemoglobin A1c (OR 2.47, 95%CI 1.17–5.21; $p = 0.018$). No other factors, such as VFA, were associated with COVID-19 severity.

3.3. Associations between clinical factors and COVID-19 outcomes

To analyze the association of clinical factors and COVID-19 outcomes, we compared clinical factors between survivors and non-survivors (Table 4), with a significantly higher proportion of patients ≥ 65 years of age among non-survivors ($p = 0.006$). The proportion of patients with dyslipidemia was significantly higher among survivors ($p = 0.025$). The proportion of patients with leukemia or lymphoma was significantly higher among non-survivors ($p = 0.0021$). The proportion of patients with serum albumin < 3.5 g/dL was significantly higher among non-survivors ($p < 0.0001$). Median VFA was significantly smaller among non-survivors ($p = 0.0003$) and the proportion of patients with VFA < 100 cm² was also significantly higher among non-survivors ($p = 0.0007$). Treatment status was not associated with outcome.

Based on the results of univariate analyses, logistic regression analysis was performed for associations between outcome and each factor (Table 5). Two factors showed an association with non-survivor status: hypoalbuminemia (OR 8.38, 95%CI 2.37–29.59; $p = 0.0010$) and low VFA (OR 3.60, 95%CI 1.15–10.06; $p = 0.027$). No other factors were associated with outcome in multivariate analysis.

As hypoalbuminemia and low VFA were independently associated with mortality, a decision tree was created and mortality rates were calculated using cutoffs of 3.5 g/dL for serum albumin and 100 cm² for VFA (Fig. 2). Patients with VFA < 100 cm² and serum albumin < 3.5 g/dL showed a higher mortality rate (38.0 %) than the other groups; 1.3 % for VFA ≥ 100 cm² and serum albumin ≥ 3.5 g/dL, 5.5 % for VFA < 100 cm² and serum albumin ≥ 3.5 g/dL, and 14.3 % for VFA ≥ 100 cm² and serum albumin < 3.5 g/dL, respectively, and the χ^2 test showed a significant difference against all other groups ($p < 0.01$). An association was found between the results and outcome of classification for patients classified into the four groups using cutoffs of 100 cm² for VFA and 3.5 g/dL for serum albumin ($p < 0.0001$, χ^2 test).

4. Discussion

The purpose of this study was to evaluate associations between clinical factors (including VFA) and COVID-19 severity and COVID-19 mortality among patients with COVID-19 during the Omicron variant epidemic.

VFA was not associated with COVID-19 severity. In a study conducted before the Omicron variant epidemic, high VFA was associated with COVID-19 severity [9]. When classifying VFA using a cutoff value of 100 cm², the group with higher VFA showed significantly higher severity than the group with lower VFA [9].

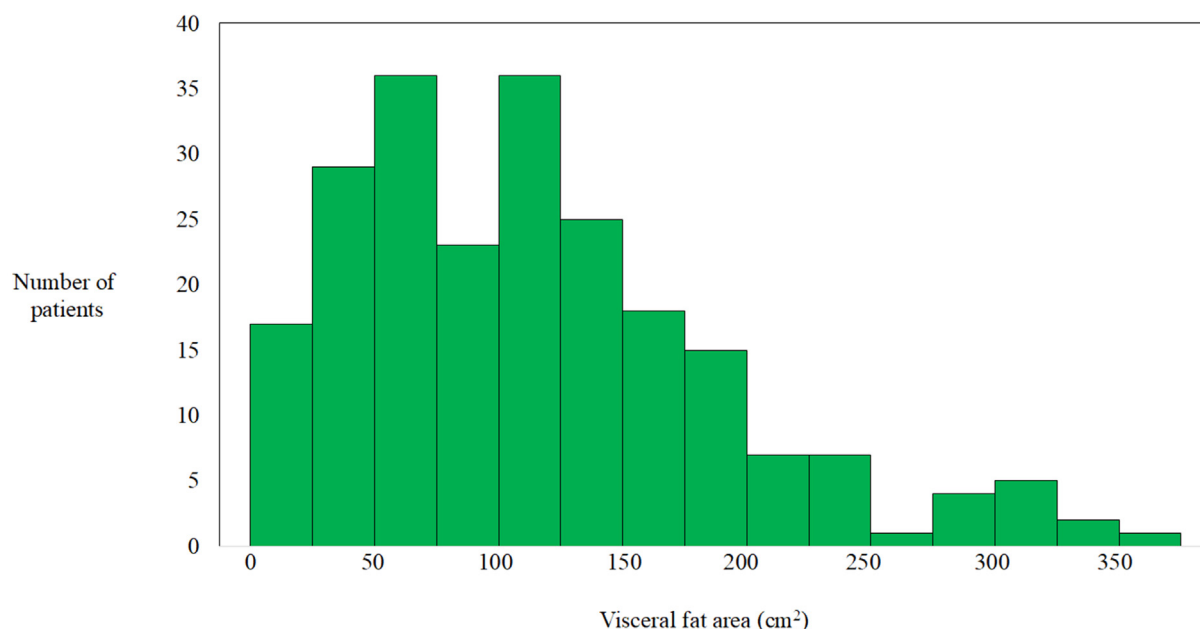


Fig. 1. Histogram of visceral fat area in subject patients.

Table 2

Comparison of clinical factors between mild and moderate-severe groups of hospitalized patients with COVID-19.

Clinical factor	Mild (n = 160)		Moderate-severe (n = 66)		p
Age (years)^a	71.5	22–97	77.5	24–92	0.013
Age ≥65 years (n, %)	93	(58.1)	50	(75.8)	0.012
Male (n, %)	87	(54.4)	46	(69.7)	0.033
Comorbidities (n, %)					
Hypertension	81	(50.6)	38	(57.6)	0.34
Dyslipidemia	53	(33.1)	19	(28.8)	0.52
Coronary artery disease or heart failure	25	(15.6)	16	(24.2)	0.13
COPD or bronchial asthma	12	(7.5)	9	(13.6)	0.15
Solid tumors	39	(24.4)	11	(16.7)	0.20
Lymphomas and leukemia	11	(6.9)	1	(1.5)	0.10
Hemodialysis	6	(3.8)	2	(3.0)	0.80
Laboratory data					
Serum albumin (g/dL) ^a	3.8	1.4–4.8	3.2	1.4–5.3	<0.0001
Serum albumin <3.5 g/dL (n, %)	51	(31.9)	41	(62.1)	<0.0001
Hemoglobin A1c (%) ^a	5.9	4.6–12.5	6.2	4.9–14.2	0.0010
Hemoglobin A1c ≥ 6.5 % (n, %)	31	(20.3)	24	(41.3)	0.0018
eGFR (mL/min/1.73 m ²) ^a	58.6	3.0–136.3	48.2	8.4–114.7	0.039
eGFR <60 mL/min/1.73 m ² (n, %)	86	(53.8)	41	(62.1)	0.25
Visceral fat area (cm ²) ^a	103.0	5.3–354.5	115.2	7.57–313.0	0.71
Visceral fat area <100 cm ² (n, %)	77	(48.1)	28	(42.7)	0.43

Abbreviations: COPD, chronic obstructive pulmonary disease; eGFR, estimated glomerular filtration rate. Data represent the number of patients, with percentages in parentheses.

^a Age, serum albumin, hemoglobin A1c, eGFR, and visceral fat area are given as median and range for each value.

Although comorbidities were present in 58.7 % of all patients and 68.9 % of critically ill patients in that previous study [9], 85.8 % of patients had some form of comorbidity in the present study. We speculate that the discrepancies between findings from these

Table 3

Multivariate analysis of clinical factors associated with COVID-19 severity.

Clinical factor	Odds ratio	95 % confidence interval	p
Age ≥65 years old	1.74	0.75–4.05	0.20
Sex, male	1.34	0.66–2.72	0.41
Serum albumin <3.5 g/dL	3.93	1.91–8.07	0.0002
Hemoglobin A1c ≥ 6.5 %	2.47	1.17–5.21	0.018
Visceral fat area <100 cm ²	0.94	0.45–2.00	0.88

studies in terms of the role of VFA in the severity of COVID-19 were attributable to differences in the susceptibility to pneumonia between patients with the Omicron variant and those with previous variants, differences in risk of comorbid disease complications as triggers for pneumonia, and changes in access to medical care in the intervening time. No reports have examined associations between VFA and severity after the Omicron variant outbreak. The present study indicates that the effect of VFA on COVID-19 severity may have changed during the Omicron variant epidemic, but further analyses are needed to verify the association between visceral fat volume and conventional severity.

On the other hand, lower VFA was associated with increased mortality in the present investigation. A study conducted before the Omicron variant epidemic found that VFA was not associated with

Table 4

Comparison of clinical factors between surviving and non-surviving hospitalized patients with COVID-19.

Clinical factor	Survivors (n = 197)		Non-survivors (n = 29)		p
Age (years)^a	72	22–97	83	24–92	0.0018
Age ≥65 years (n, %)	118	(59.9)	25	(86.2)	0.006
Male (n, %)	86	(56.9)	21	(72.4)	0.11
Clinical background (n, %)					
Hypertension	107	(54.3)	12	(41.4)	0.19
Dyslipidemia	68	(34.5)	4	(13.8)	0.025
Coronary artery disease or heart failure	34	(17.3)	7	(24.1)	0.37
COPD or bronchial asthma	20	(10.2)	1	(3.5)	0.25
Solid tumor	44	(22.3)	6	(20.7)	0.84
Lymphoma or leukemia	7	(3.6)	5	(17.2)	0.0021
Hemodialysis	8	(4.1)	0	(0)	0.27
Laboratory data at admission or diagnosis					
Serum albumin (g/dL) ^a	3.8	1.4–4.8	2.6	1.4–5.3	<0.0001
Serum albumin <3.5 g/dL (n, %)	67	(34.0)	25	(86.2)	<0.0001
Hemoglobin A1c (%) ^a	6	4.6–14.2	5.7	5.1–12.8	0.27
Hemoglobin A1c ≥ 6.5 % (n, %)	51	(26.7)	5	(20.8)	0.54
eGFR (mL/min/1.73 m ²) ^a	57.8	3.0–136.3	46.9	8.4–115.8	0.066
eGFR <60 mL/min/1.73 m ² (n, %)	108	(54.8)	19	(65.5)	0.29
Visceral fat area (cm ²) ^a	114.6	6.2–354.5	55.7	5.3–254.2	0.0003
Visceral fat area <100 cm ² (n, %)	83	(42.1)	22	(75.9)	0.0007
Medical treatment (n, %)					
Treated	162	(82.2)	23	(79.3)	0.70
Remdesivir	124	(62.9)	22	(75.9)	0.17
Dexamethasone	47	(23.9)	12	(41.4)	0.044

Abbreviations: COPD, chronic obstructive pulmonary disease; eGFR, estimated glomerular filtration rate. Data represent the number of patients, with percentages in parentheses.

^a Age, serum albumin, hemoglobin A1c, eGFR, and visceral fat area are given as median and range for each value.

Table 5

Multivariate analysis of clinical factors associated with mortality in hospitalized patients with COVID-19.

Clinical factor	Odds ratio	95 % confidence interval	p
Age ≥65 years (n, %)	1.75	0.47–6.46	0.54
Dyslipidemia	0.46	0.11–1.87	0.28
Lymphoma or leukemia	1.97	0.43–9.02	0.38
Serum albumin <3.5 g/dL	8.38	1.90–29.59	0.0010
Visceral fat area <100 cm ²	3.40	1.15–10.06	0.027
Remdesivir used	1.46	0.49–4.41	0.50
Dexamethasone used	1.03	0.32–3.26	0.96

increased mortality within 30 days [13]. Median VFA in that previous study was 180.6 cm², and 21.4 % of patients had a VFA ≤100 cm² [13]. In comparison, median visceral fat tissue in the

present study was 108.4 cm², and 46.5 % of patients had a VFA <100 cm², indicating that many patients in the population had low VFA. This suggests that the high mortality rate was strongly influenced by the proportion with extremely low VFA. Previous reports have shown a J-shaped, nonlinear relationship between body mass index (BMI) and COVID-19 mortality, with a flat dose-response curve for BMI around 22–24 kg/m² and increased mortality for lower BMIs [14]. BMI correlated positively with VFA [15], suggesting that low VFA may be associated with increased mortality in COVID-19. Adipose tissue contains immune cells such as macrophages, lymphocytes, and neutrophils, and contributes to innate immunity by promoting macrophage migration [16,17]. Adipose tissue promotes the secretion of tumor necrosis factor α (TNF- α) [18], and the use of TNF- α inhibitors has been associated with an increased risk of severe infections [19]. Low VFA may reduce innate immune

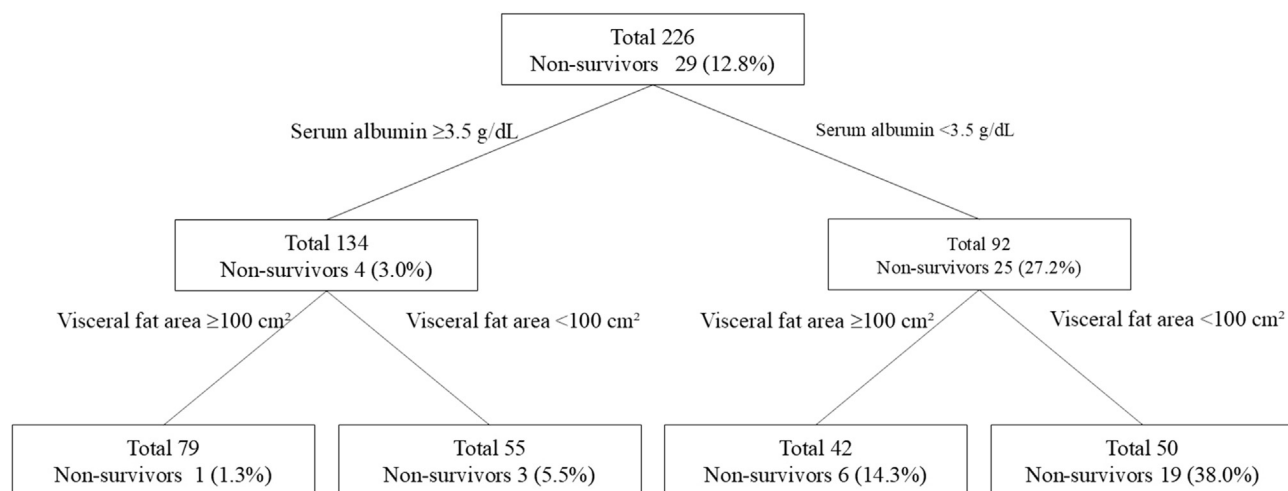


Fig. 2. Decision tree analysis of patient mortality rates classified by serum albumin level and visceral fat area. Patients were initially classified with a cutoff value of 3.5 g/dL for serum albumin, then classified into four groups using a cutoff value of 100 cm² for visceral fat area.

function, which may put the patient at risk of COVID-19 mortality. Cachexia, which decreases muscle and fat mass, is progressively debilitating and can accelerate mortality not only from cancer but also from infection [20]. Cachexia also results in elevated levels of inflammatory cytokines [20]. Although it cannot be determined whether all deaths in this study were due to cachexia, it is possible that patients with low VFA and elevated inflammatory cytokines, as well as cachexia, may have contributed to the observed mortality. Further studies might be needed to clarify the effects of adipose tissue on the innate immune system in COVID-19. In addition, the association between low body fat mass and increased mortality in the elderly with chronic disease, has been known as obesity paradox [21]. To determine whether this association in COVID-19 patients was due to nonspecific phenomenon, comparisons between COVID-19 and other viral infections would be needed.

Hypoalbuminemia in association with metabolic factors other than VFA was associated with both more severe COVID-19 and increased mortality. Low serum albumin levels have been reported in previous meta-analyses as a risk factor for COVID-19 severity and mortality [22]. In another report, serum albumin was significantly lower in patients who required invasive mechanical ventilation on admission compared with those who did not, and serum albumin was significantly lower in patients who died within 90 days of admission compared to those who survived [23]. No specific serum albumin cutoff has been established for COVID-19, and even in meta-analyses where hypoalbuminemia has been identified as an adverse prognostic factor for COVID-19, cutoff values have varied across different studies [24]. The cause of association between hypoalbuminemia and increased severity and mortality of COVID-19 may be that COVID-19 itself has resulted in decrease in serum albumin. Serum albumin levels are decreased during the acute phase response associated with acute and chronic illness and inflammation [25]. In COVID-19, the increased inflammatory response caused by the release of numerous proinflammatory cytokines such as interleukin-6 (IL-6), IL-1, and tumor necrosis factor- α correlates with disease severity [26]. Since the present study included the condition at the onset of COVID-19 in the analysis, changes in serum albumin from before the onset cannot be clarified, and studies tracking serum albumin levels prior to the onset would be needed. Hypoalbuminemia is a mortality prognostic factor as well as an indicator of malnutrition in the elderly regardless of comorbidities [27]. In the present study, 63.3 % of patients were over 65 years of age, and hypoalbuminemia was an independent risk factor, as in previous reports [10]. The relationship between hypoalbuminemia and increased mortality in COVID-19 is speculated to be multifactorial [23]. Although there may be a single factor that causes hypoalbuminemia, we believe that albumin is a routinely measured value that, in combination with other factors, provides additional predictive accuracy for COVID-19 prognosis.

For hypoalbuminemia and low VFA, which were identified as independent mortality-related factors, a decision tree was created and mortality rates were calculated using cutoffs of 3.5 g/dL for serum albumin and 100 cm² for VFA. Patients with VFA <100 cm² and serum albumin <3.5 g/dL had a significantly higher mortality rate than patients in other groups. No previous reports have evaluated patient mortality using the two factors of VFA and serum albumin. Although several reports have examined prognostic factors for COVID-19, various endpoints have been examined and no consensus has yet been reached [28]. Risk of death from COVID-19 could potentially be easily estimated using VFA and serum albumin. By predicting mortality risk using serum albumin levels and VFA at the time of hospitalization, it may be possible to consider the proactive antiviral therapy for COVID-19 patients with hypoalbuminemia and low VFA, even if they were conventionally assessed as having a low risk of severe disease.

The present study has several limitations. First, our study was a single-center study with a small sample size and included some patients who were infected during hospitalization, which may have shifted the patient background from the community population of COVID-19-infected patients. Our study focused on hospitalized patients with COVID-19, which may have overestimated the risk of mortality compared to non-hospitalized patients. However, the mortality rate for hospitalized patients approximated that seen from domestic data, and we believed that the risk assessment was therefore reasonable. We considered that the results of this study can be generalized by conducting future multicenter studies and increasing the sample size. Second, comorbidities were determined based on past medical records and medication status, but comorbidities not treated with medication may have been underestimated and not collected. However, 85.8 % of patients had some comorbidities, and we believe that we were able to assess comorbidities appropriately, at least to some extent. Third, all of patients in this study were Asian, but it is unclear whether the results of this study apply to other races. Ethnic differences in COVID-19 mortality rates were reported in the early stages of the COVID-19 epidemic, but have decreased with the epidemic of the Omicron variant [29]. It is necessary to investigate whether similar results will be observed in other races and nationalities, but we can expect similar outcomes. Fourth, the present study could not fully eliminate confounding factors due to lack of data on novel coronavirus vaccine status and history of COVID-19 infection in the retrospective analysis. The impact of COVID-19 vaccination or reinfection on COVID-19 severity and mortality varies among reports [30–33]. Prospective studies that include vaccination status with novel coronavirus vaccines and history of COVID-19 in the analysis will provide results that are relevant to the current infection situation, in which opportunities for re-infection are increasing.

In conclusion, low visceral fat volume and hypoalbuminemia were associated with mortality among hospitalized patients with COVID-19 during the omicron variant epidemic. There were no reports evaluating the mortality rate of COVID-19 based on VFA and serum albumin. We suggested that setting cutoff values for VFA by abdominal CT and serum albumin may allow simple prediction of mortality risk among hospitalized patients with COVID-19.

Author contributions

Shin Nakayama: Data curation; Formal analysis, Investigation, Methodology, Software, Writing - original draft, Visualization. This research did not receive any specific grant from funding agencies in the public, commercial, or no-for-profit sectors.

Yoshitaka Wakabayashi: Formal analysis, Investigation, Writing - review & editing.

Kyotaro Kawase: Formal analysis, Investigation.

Ai Yamamoto: Formal analysis, Investigation.

Takatoshi Kitazawa: Conceptualization, Formal analysis, Funding acquisition, Investigation, Methodology, Project administration, Resources, Supervision, Validation, Visualization, Writing - review & editing.

Declaration of competing interest

The authors declare that they have no competing interests.

Acknowledgments

We thank Forte Science Communications for language assistance.

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